
PH 505: Electronic Structure

Course Code	PH 505
Course Name	Electronic Structure
L-T-P-C	3-0-0-3
Approval	2nd Adhoc Senate (May 2012), in the Minor in Applied Physics curriculum

Course Contents

- **Overview**
- Quantum theory and origin of electronic structure, electronic ground state, basic equations for interacting electron and nuclei, periodic solids and bands, uniform electron gas and simple metals. [7-8 Lectures]
- **Density functional theory**
- DFT foundations, Thomas Fermi Dirac approximations, Hohenberg-Kohn Theorems, intricacies of DFT, Kahn Sham variational equations, Time dependent DFT, local spin density approximation, GGA, LDA, saving Kahn-Sham equations [7-8 Lectures]
- **Important preliminaries on atoms**
- One electron Schrodinger equation, relativistic Dirac equation, atomic sphere approximations, pseudopotentials, orthogonalized plane waves, ultrasoft potentials, projected augmented waves. [7-8 Lectures]
- **Determination of Electronic structure**
- Bloch Theorem, Nearly free electron model, ab initio pseudopotential method, crystal structure, supercells, clusters and molecules, tight binding methods, augmented functions: APw, MTO, linear methods, LAPW. [7-8 Lectures]
- **Module V**
- Predicting properties of matter from electronic structure - recent developments and computational resources in use. [7-8 Lectures]

Text Books

1. R.M. Martin, Electronic Structure: **Basic theory and practical methods**, Cambridge University Press, 2004.

References

1. R.M. Martin, Electronic Structure: **Basic theory and practical methods**, Cambridge University Press, 2004.
2. Ashcroft and Mermin, **Solid State Physics**, Holt, Rinehart and Winston, 1976.
3. Kittel, **Introduction to Solid State Physics**, Wiley, 1986.
4. Omar, **Elementary Solid State Physics**, Addison-Wesley, 1975.
5. Ziman, **Principles of the Theory of Solids**, Cambridge, 1972.
6. W. Hergert A. Ernst M. D " ane (Eds.), **Computational Materials Science: From Basic Principles to Material Properties**
7. JMD Coey, **Magnetic Materials**